# WEB PROGRAMMING

## Midterm Project — Pitch Presentation Framework

**PROJECT TITLE:**

Team Name: ________________________________

Date: ________________________________

**TEAM MEMBERS**

| # | Full Name | Student ID | Role in Project |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

**Duration: 10–15 min | Total: 100 Points | Dynamic Web Projects Only**

## How to Use This Document

This document is your **step-by-step framework** for planning and delivering your midterm pitch. Every project must be a **dynamic web application** that solves a real problem for real users  not a simple static page.

Fill in every grey box before your presentation day. Bring this printed document as your speaker notes.

| Section | Slide / Topic | Time | Points |
|---|---|---|---|
| 1 | Introduction | 1–2 min | 10 |
| 2 | What Is the Web App? | 2 min | 15 |
| 3 | Framework Used | 2 min | 15 |
| 4 | Technologies Used | 2–3 min | 20 |

| | | | |
|---|---|---|---|
| 5 | Technology Overview Diagram | 2 min | 20 |
| 6 | Project Overview | 3–4 min | 20 |

## Overall Presentation Flowchart

Use this flowchart as your preparation checklist and as a visual guide on presentation day. Each numbered box is one section of your pitch. Complete them in order  top to bottom.

### PRESENTATION FRAMEWORK — OVERALL FLOWCHART

*Follow this sequence from top to bottom for your midterm pitch*

**▶ START PRESENTATION**

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**1 …**

**INTRODUCTION**

- Greet the audience and introduce yourself
- State your project name and one-line pitch
- Describe the real-world problem you are solving
- Identify who is affected (your target user)

**10 pts**
1–2 min

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**2 …**

**WHAT IS THE WEBSITE / APP?**

- State the type: Dynamic Web  / E-Commerce / Social Platform
- Every project must solve a real problem for real users
- Describe your 3 key features use action verbs (search, book, track…)
- Show a screenshot or wireframe of the homepage

**15 pts**
2 min

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**3 …**

**FRAMEWORK USED**

- Name your framework (React / Vue / Bootstrap / Laravel / Django…)
- Is it Frontend, Backend, or Full-Stack?
- Explain what it does in the context of YOUR project
- State one alternative you considered and why you chose yours instead

**15 pts**
2 min

▼

**4 …**

**TECHNOLOGIES USED**

- Frontend Layer HTML5, CSS3, JavaScript, UI framework
- Backend Layer server language / runtime / framework

**20 pts**
2–3 min

– Database — what data is stored and how
– APIs & External Services third-party integrations
– Each technology must be explained in context, not just listed

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| | | |
|---|---|---|
| **5** ••• | **TECHNOLOGY OVERVIEW DIAGRAM**<br>– Draw a layered architecture diagram on your slide<br>– Show: Browser → Frontend → Backend → Database / API<br>– Use arrows to show direction of data flow<br>– Point to each component and explain it verbally | **20 pts**<br>2 min |

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| | | |
|---|---|---|
| **6** ••• | **PROJECT OVERVIEW**<br>– Walk through the user journey step by step<br>– Show the site map: list all frame of pages and their purpose | **20 pts**<br>3–4 min |

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**Q & A — Questions from the Instructor** *(allow 3–5 minutes)*

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**■ END**

## Section 1 — Introduction (10 pts)

The introduction is your **first impression**. You have 60–90 seconds to make the audience care. Introduce yourself, state your project name, and in plain language explain what problem your web app solves.

> **TIP:** Open with a real situation — not technology. Example: "Imagine you are a new exchange student in Taichung. You do not speak Chinese. You need to find a halal restaurant, but every app gives you 50 unverified results…" Then reveal your solution.

### 1.1 Who Are You?

> *My name is ___________ and my student ID is ___________.*

### 1.2 Project Name and One-Line Pitch

*Formula:* "[Project Name] is a web app that helps [target users] to [solve what problem]."

> *Project name:* ________________________________________________

*One-line pitch:* ______________________________________________

### 1.3 The Problem You Are Solving

Describe the real-world problem. Who is affected and why does it matter? Be specific — 2 to 3 sentences.

*The problem:* ______________________________________________
*Who is affected:* ______________________________________________
*Why existing solutions are not good enough:* ____________________________

**TIP:** Scoring tip — "People have trouble finding information online" earns 0 marks. "Exchange students in Taichung cannot find halal-certified restaurants near campus because no verified source exists in English" earns full marks.

## Section 2 — What Is the Web App? (15 pts)

**All projects must be dynamic web applications** that solve a specific problem. Explain what users can do, who they are, and how the app makes their life better.

### 2.1 Type of Dynamic Web Application

Select the type that best describes your project. Every option requires a backend and a database.

| App Type | Description | Must Have |
|---|---|---|
| Problem-Solving Web App | Users interact with the app to solve a specific daily problem — booking, searching, tracking, managing. | Login, database, dynamic content |
| E-Commerce Platform | Users browse, filter, and purchase products or services online. | Cart, payment or booking, order history |
| Social / Community Platform | Users create profiles, post content, and interact with each other. | Profiles, posts, user-to-user interaction |
| Information & Service Portal | Users register and access personalised information or services — health, education, government. | User accounts, personalised dashboard |

*My project type is:* ____________ *because it solves:* __________________

### 2.2 Target User Be Specific

**Do not write "everyone".** Describe a real group of people with a specific need.

*Example:* "First-year exchange students at FCU who need to find halal-certified food near campus."

*My target user:* ______________________________________________
*Their specific problem:* ____________________________________________
*How my app helps them:* ____________________________________________

### 2.3 Three Key Features

Each feature must directly address the user's problem. Use action verbs: search, book, upload, review, track, filter, compare, receive, send, manage.

*Feature 1:* ____________________________________________

*Feature 2:* ____________________________________________

*Feature 3:* ____________________________________________

**TIP:** Every feature must connect to the problem. Feature 1 solves the core need. Feature 2 adds convenience. Feature 3 differentiates your app from alternatives.

## Section 3 — Framework Used (15 pts)

A framework is a pre-built code structure that organises your project and speeds up development. Name your framework, explain its role in your project, and justify why you chose it over alternatives.

### 3.1 Framework Reference

| Framework | Type | Language | Best For |
|---|---|---|---|
| Bootstrap | CSS Framework | CSS + JS | Fast responsive layouts, consistent UI components across all pages |
| Tailwind CSS | CSS Framework | CSS | Custom visual designs using utility class names |
| React | Frontend Library | JavaScript | Dynamic single-page apps, reusable UI components, real-time updates |
| Vue.js | Frontend Framework | JavaScript | Beginner-friendly interactive UIs, two-way data binding |
| Next.js | Full-Stack Framework | JavaScript | React apps with server-side rendering and built-in routing |
| Express.js | Backend Framework | JavaScript (Node) | Building REST APIs, routing, middleware, server logic |
| Laravel | Backend Framework | PHP | Full MVC web apps, authentication, ORM database access |
| Django | Backend Framework | Python | Data-heavy apps, admin panel, ORM, rapid development |
| Flask | Backend Framework | Python | Lightweight APIs, simple server routing |

### 3.2 Your Framework Answer

*The framework I used is: ________________________________________________*
*It is a (circle): Frontend / Backend / Full-Stack / CSS-only framework*
*In my project it is responsible for: ________________________________________*
*Alternative I considered: ____________ I chose mine because: ___________________*

**TIP:** High-scoring answer: "I chose React because my app has a live search feature that filters results as the user types — React's component re-rendering handles this without reloading the page, which a plain HTML page cannot do efficiently."

## Section 4 — Technologies Used (20 pts)

A dynamic web app uses multiple technologies that work together as layers. You must identify what technology you used at each layer and explain what it does in your specific project.

### 4.1 The Four Layers of a Dynamic Web App

| Layer | What It Does | Common Technologies |
|---|---|---|
| Frontend — Presentation | What the user sees: layout, styles, interactive behaviour in the browser. | HTML5, CSS3, JavaScript, React, Vue, Bootstrap, Tailwind |
| Backend — Logic | The server that receives requests, runs business logic, and sends responses. Required for user accounts, data processing, security. | Node.js + Express, PHP + Laravel, Python + Django or Flask |
| Database — Storage | Stores all persistent data: user accounts, posts, bookings, products, messages. | MySQL, PostgreSQL, MongoDB, Firebase Realtime DB, SQLite |
| External APIs | Third-party services your app calls to get data or add functionality. | Google Maps, Stripe, Firebase Auth, OpenWeather, Cloudinary |

### 4.2 Your Technology Stack — Fill In

| Layer | Technology You Used | What You Use It For in Your Project |
|---|---|---|
| Frontend — Structure | | |
| Frontend — Styling | | |
| Frontend — Interactivity | | |
| Backend / Server | | |
| Database | | |

| | | |
|---|---|---|
| Authentication | | |
| External API or Service | | |
| Hosting / Deployment | | |

### 4.3 Explain Each Technology in Context

*Write one sentence per technology explaining its role in YOUR project — not a dictionary definition.*

*Example:* "I use MySQL to store each restaurant's name, address, halal certification status, and average rating. Every time a user searches, my backend queries this table and returns matching records."

*Technology 1 — Name:* _____________ *— In my project it:* ________________________
*Technology 2 — Name:* _____________ *— In my project it:* ________________________
*Technology 3 — Name:* _____________ *— In my project it:* ________________________
*Technology 4 — Name:* _____________ *— In my project it:* ________________________

## Section 5 — Technology Overview Diagram (20 pts)

An architecture diagram is a **visual map** of your entire system — showing every component and how data flows between them. You must draw this diagram on a slide and explain every part verbally.

### 5.1 What the Diagram Must Show

- The browser or client device (how the user accesses the app)
- The frontend layer — HTML, CSS, JavaScript, and your UI framework
- HTTP / HTTPS communication arrow between browser and server
- The backend server — the language, runtime, and framework
- API call arrows between server and database / external services
- The database — what data is stored there
- Any external APIs or services your app integrates

### 5.2 Standard Architecture for a Dynamic Web App

### 5.3 Your Diagram Sketch — Plan It Here

[ Draw or paste your architecture diagram here ]

### 5.4 Explain Every Arrow

For every arrow in your diagram you must state: (1) what component sends the data, (2) what component receives it, and (3) what data is travelling.

**TIP:** Practise pointing to your diagram and saying each arrow out loud. Example: "This arrow goes from the browser to my Express server — it carries the user's search keyword as a query string in the URL."

## Section 6 — Project Overview (20 pts)

The overview of the project how it will look like .

## Grader Checklist

| Section | Full-Marks Criteria | Pts |
|---|---|---|
| Introduction | Clear project name, one-line pitch, specific problem, named target user | 10 |
| What Is the App? | Dynamic type stated, specific user identified, 3 features described with action verbs | 15 |
| Framework | Named, type identified, role explained in project context, alternative considered | 15 |
| Technologies Used | All 4 layers covered, each tech explained in project context, not just listed | 20 |
| Architecture Diagram | Diagram shown on slide, every component named, every arrow explained verbally | 20 |
| Overview | Overview of the website how it will look like what would be the core features | 20 |
| Total | | 100 |

## Quick Glossary

## Notes